

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1718

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 Total Marks:30

Annexure A sDry Run

Evaluation

Code of Conduct:

I **A . Bharath / 192512384**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A . Bharath

1. Breadth-First Search (BFS)

Graph

A
/ \
B C
/ \
D E F G

Table

Iteration	Queue	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Answers:

1. **BFS Order:** $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$
2. **Queue:** $B, C \rightarrow C, D, E \rightarrow D, E, F, G \rightarrow E, F, G \rightarrow F, G \rightarrow G \rightarrow \text{Empty}$

2. Depth-First Search (DFS)

Table

Iteration	Stack	Visited Node
1	C, B	A
2	C, E, D	B

3	C, E D
4	C E
5	G, F C
6	G F
7	Empty G

Answer:

- DFS Order: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$
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3. 8-Puzzle (BFS)

Answers

1. Expanded Node (First 5 Iterations)

Iteration Expanded Node

1	Initial State
2	First Successor
3	Second Successor
4	Third Successor
5	Fourth Successor

2. Total States Generated

- Around **15–20** (first five iterations).

3. Solution Path

- Initial State $\rightarrow \dots \rightarrow$ Goal State (found after complete BFS).

4. Why BFS Finds the Shortest Path?

- BFS explores **level by level**, so the first goal found is the **shortest path**.